(c) What are linear block codes? What is matrix representation of block codes. 7

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(d) Explain majority logic (Threshold) decoding with neat diagram and table. 7

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B. E. (Sixth Semester) Examination, Nov.-Dec., 2015

(Old Scheme)

(ET&T Engg. Branch)

COMMUNICATION SYSTEM-II

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all the questions. Part (a) of each part is compulsory. Attempt any two from (b), (c) and (d).*

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Unit-I

1. (a) What is Nyquist rate and Nyquist interval. 2

(b) State and prove sampling theorem in time domain. 7

(c) What is "frequency division multiplexing"?Draw the block diagram of FDM system and explain its working. 7

(d) Explain generation of PPM and PWM with the help of block diagram and waveforms. 7

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Unit-II

2. (a) Define Quantization. 2

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(b) Draw the block diagram of Adaptive Delta modulation and explain its working. 7

(c) Draw the functional block diagram of differential pulse code modulation and explain the function of each block. 7

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(d) What is companding? Explain how companding improves the signal to noise ratio of a PCM system. 7

Unit-III

3. (a) Sketch the curve for power spectral density of binary PSK. 2

(b) Draw the functional block diagram of QPSK system and explain its working. 7

(c) Draw the block diagram of DPSK modulator and explain how synchronization problem is avoided for its detection. 7

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(d) Derive an expression of probability of error for QPSK. 7

Unit-IV

4. (a) Write an expression of signal to quantization noise ratio for pulse code modulation. 2

(b) Draw the functional block diagram of direct sequence and explain its working spread spectrum. 7

(c) Explain working of frequency hopped spread spectrum system with suitable diagram. 7

(d) Explain the principle of CDMA list important advantages of CDMA. 7

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Unit-V

5. (a) Define entropy and channel capacity related with information theory. 2

(b) Explain Shanon Fano coding with the help of example. 7